

Ludvig Fellstrom

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EDUCATION

Cornell University, College of Engineering, Ithaca, NY
Bachelor of Science, Electrical and Computer Engineering
GPA: 3.6

Expected May 2028

Relevant Courses: Computer Architecture, Foundations of Machine Learning, Algorithms & Data Structures

PROFESSIONAL EXPERIENCE

IBM, Austin, TX, *Power Systems Development Intern*

May 2026 - Present

- Developed an autonomous diagnostics platform combining YOLO-based physical inspection with Linux-on-Power debugging for PCIe, device configuration, and firmware redundancy checks.
- Developed Rust-based DevOps tooling with Grafana and Prometheus for node telemetry and LPAR utilization monitoring.
- Supported cross-stack PyTorch enablement for IBM Spyre accelerators and GPU compatibility on Power systems.

Ghost Social, San Francisco, CA, *AI Engineering Intern*

May 2025 - Dec 2025

- Built Python developer tooling and agent-driven CI/CD to automate and monitor local deployments and surface code review to detect integration issues earlier in the development workflow.
- Supported production backend infrastructure in Python, including PostgreSQL-backed workflows, Git-based CI/CD, and containerized deployment.

RESEARCH EXPERIENCE AND PROJECTS

FPGA Packet Router, Cornell High Frequency Trading Club, *Founding Team Member*

Jan 2026 - Present

- Designed an FPGA packet router in SystemVerilog that prioritized freshest packets through timestamp comparison and destination-based forwarding logic.
- Wrote SystemVerilog Assertions and cocotb testbenches to verify routing correctness, freshness properties, and bounded edge-case behavior.

Progressive Localization via Optimal Transport, Goldfeld Lab, *Undergraduate Researcher*

July 2026 - Present

- Supported the development of scalable pipelines for activation analysis and causal intervention experiments to study how transformers represent and use information.
- Applied optimal transport and subspace methods to connect interpretable reasoning steps with distributed neural representations.

TECHNICAL EXPERIENCE

Cornell Autonomous Sailboat, Cornell University, *Navigation Subteam Member*

Aug 2024 - Present

- Aided in the development of a PyTorch YOLO pipeline on Jetson Orin Nano using CUDA, Docker, and ROS 2, achieving 93% mAP.
- Headed software and electrical integration, leading to a 1st place win in the International Robotic Sailing Regatta.

ECE 2400 Computer Systems Programming, Cornell University, *Teaching Assistant*

Jan 2026 - May 2026

- Developed a Rust-based coding environment with a course-constrained AI tutor, enabling guided assistance while preserving lab integrity.
- Led office hours on dynamic memory management, pointer aliasing, and C data structures for 80+ students.

TECHNICAL SKILLS

Languages: Python, C/C++, Java, SystemVerilog, Verilog, Bash, Assembly (RISC-V)

Tools/Frameworks: ROS 2, CUDA, TensorRT, PyTorch, Spring, Docker, Git, PostgreSQL, Grafana, Kubernetes, cocotb